

(BALUCHISTAN PUBLIC SERVICE COMMISSION)

# I.T TEACHER

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**BALUCHISTAN PUBLIC SERVICE COMMISSION**  
WRITTEN EVALUATION TEST FOR RECRUITMENT TO THE POST(S) OF  
**LECTURER (B-17)**  
IN THE GOVERNMENT COMMERCE COLLEGE EDUCATION DEPARTMENT  
**COMPUTER SCIENCE**

Roll Number

Total Marks: 100

February 20, 2019

Time Allowed: 03 Hours

- NOTE:**
- Attempt all questions.
  - All the parts (if any) of each Question must be attempted at one place instead of at different places.
  - Candidate must write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
  - No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.

[Objective: 40 Marks]

**Q 1-A: Fill in the blanks: (C - Language)**

(05 Marks)

- Sequences such as \t, \n, \r are called \_\_\_\_\_.
- To store a string such as "Computer Science" an array of data type \_\_\_\_\_ is required.
- Data type that stores address of a variable is \_\_\_\_\_.
- A composite data type that can store a group of variable is \_\_\_\_\_.
- A loop that will execute at least once despite the condition is \_\_\_\_\_.

**Q 1-B: Choose the correct option. (Object Oriented Programming)**

(05 Marks)

- A constructor has the same name as that of  
a) member function b) object c) member data d) class
- A special function/method used to initialize instance of a class  
a) Member function b) Destructor c) Constructor d) initializer
- Which of the following is not feature of object oriented programming(OOP)?  
a) Code reusability b) Modularity c) Duplicate data d) Efficient code
- Which feature of OOP illustrate code reusability  
a) Polymorphism b) Abstraction c) Encapsulation d) Inheritance
- Encapsulation can be achieved using ?  
a) access specifiers b) only private members c) inheritance d) Abstraction

**Q 1-C: Fill in the blanks. (Database Management Systems)**

(05 Marks)

- A collection of related data is called \_\_\_\_\_.
- A software that manages and controls access to database is called \_\_\_\_\_.
- A program that interacts with database at some point in its execution is called \_\_\_\_\_.
- A key that serves to uniquely identify records in a table is \_\_\_\_\_.
- A language that allows users to insert, update, delete and retrieve data from the database is \_\_\_\_\_.

**Q 1-D: Fill in the blanks. (Data Structures and Algorithms)**

(05 Marks)

- Name of the searching algorithm that requires data to be sorted \_\_\_\_\_.
- Name of the data structure suitable for storing printing jobs in FIFO style \_\_\_\_\_.
- Name of the data structure suitable for storing elements in LIFO style \_\_\_\_\_.
- Name of the sorting algorithm that does not compare elements \_\_\_\_\_.
- A tree that has at most two children is called \_\_\_\_\_.

**Q 1-E: Fill in the blanks (Compiler Construction)**

(05 Marks)

- Compiler usually reports the presence of \_\_\_\_\_ in source program.
- Compiler translate source code to \_\_\_\_\_.
- A grammar that produces more than one parse tree is \_\_\_\_\_.
- Lexical analysis breaks a sequence of characters into \_\_\_\_\_.
- A data structure which records the information related to identifiers used in a source program is \_\_\_\_\_.

**Q 1-F: Fill in the blanks. (Network & Operating System Concepts)**

(05 Marks)

- A popular seven layered conceptual model which standardizes communication functions is \_\_\_\_\_.
- A standard protocol used for the transfer of files between a client and server \_\_\_\_\_.
- Name of the protocol used by the world wide web \_\_\_\_\_.
- Feature of an operating system that allows more than one program to run concurrently \_\_\_\_\_.
- A key component of the operating system that connects application software to the hardware of a computer is \_\_\_\_\_.

**Q 1-G: Write complete form of the following:**

(10 Marks)

- NTFS 2. BIOS 3. RAID 4. USB 5. HDMI 6. UUID 7. W3C 8. XML 9. ROM 10. SMTP



[Subjective: 60 Marks]

(05 Marks)

Question 2: What will be the output of the following c program

```
void main(void)
{
    int num = 10;
    printf("%d\n", num--);
    printf("%d\n", ++num);
}
```

Question 3: Factorial of a number  $n$  is the product from 1 to  $n$ , e.g.  $4! = 1 \cdot 2 \cdot 3 \cdot 4 = 24$ . Write a program in C or C++ with a function named "fact" which takes an integer 'n' as argument and returns its factorial. (10 Marks)

Question 4: Suppose a table of city with columns id, cityname, population, province. Write SQL queries to achieve the following: (10 Marks)

1. Display all records in table with cityname in ascending order
2. Display all records in table with population in ascending order
3. Display only those records where province value is 'KPK'
4. Display only those records where value of population is greater than 2,000,000
5. Display sum of population grouped by province

Question 5: A school requires a database to store information regarding their teachers, students, courses, classes, sections. Draw an Entity Relationship diagram keeping in mind the following points: (10 Marks)

One class is divided into many sections, one teacher can teach many courses, each class has one teacher as class in charge

Question 6: Given the name and worst case time complexity, arrange the following algorithm according to efficiency i.e. Best algorithm first. (05 Marks)

Selection Sort:  $n^2$ , Heap Sort:  $n \log n$ , Quick Sort:  $n^2$ , Merge Sort:  $n \log n$

Question 7: Perform Radix sort on the following list of numbers

54400, 54401, 54410, 54201, 53400, 54461, 57400, 54408, 54490, 58401

(10 Marks)

Question 8: Given following context free grammar derive the arithmetic expression  $(5 - 3) * 2 + 3$

```
<expr> -> number
<expr> -> ( <expr> )
<expr> -> <expr> + <expr>
<expr> -> <expr> - <expr>
<expr> -> <expr> * <expr>
<expr> -> <expr> / <expr>
```

(10 Marks)



# BALUCHISTAN PUBLIC SERVICE COMMISSION

Samungli Road, Quetta Cantt.

Written Evaluation Test for the Post(s) of

Lecturer (B-17)

In the Colleges, Higher and Technical Education Department

April 06, 2017

Time Allowed: 03 Hours

Total Marks: 100

## Computer Science

Note: Attempt six questions. Question No. 1 is compulsory

### Question 1:

#### A) Fill in the blanks:

(10)

- i. Mesh Topology is \_\_\_\_\_ flexible and has a \_\_\_\_\_ expandability.
- ii. \_\_\_\_\_ is approximately 1,000 bytes.
- iii. Process known as \_\_\_\_\_ tracks trends and allows retailers to respond to consumer buying patterns.
- iv. The main attribute of Spiral model for software development is \_\_\_\_\_.
- v. The keyword used to transfer control from a function back to the calling function is \_\_\_\_\_.
- vi. While coding in C/C++ programming for "Call by Reference" we pass \_\_\_\_\_.
- vii. ISP exchanges internet traffic between their networks by \_\_\_\_\_.
- viii. IPv6 addresses have a size of \_\_\_\_\_.
- ix. System software are categorized in \_\_\_\_\_ and \_\_\_\_\_.
- x. The \_\_\_\_\_ contains commands associated with the My Computer window.

#### B) For whom the following abbreviation stands for:

(5)

- i. CMOS
- ii. CASE
- iii. CSMA/CD
- iv. SDLC
- v. SRS

#### C) Write "TRUE" or "FALSE":

(10)

- i. The word bit is the short form of Binary Digit.
- ii. Fourth generation computer use integrated circuits.
- iii. Binary value of 303 is 001100011.
- iv. Parallel data transmission is much faster than serial transmission.
- v. Counters are digital circuit for temporary storage of data.
- vi. Hexadecimal Value of 11 is C.
- vii. To display the contents of a file use LIST command.
- viii. DEL command deletes a directory in DOS.
- ix. Starting process of a computer is called cold booting.
- x. A set of instructions is called Software.

### Question 2:

- a) Explain the functionality of seven layers of OSI reference model and map with TCP/IP protocol suite? (5)
- b) Explain Amplitude Modulation and Frequency Modulation and where do we use AM and FM? (5)
- c) What is Proxy Server? (5)

### Question 3: Write down difference between:

- a) Data Hiding and Encapsulation (5)
- b) 3G and 4G (5)
- c) Microcontroller and microprocessor (5)

### Question 4:

- a) Draw an entity relationship diagram between EMPLOYEES, DEPARTMENTS and PROJECTS assuming that each project can be started by only one department and each employee can be employed by only one department at a time. Write down any other assumption if you use it. (5)
- b) What do you mean by atomicity and aggregation? (5)
- c) Describe the three levels of data abstraction? (5)

### Question 5:

- a) Which data structure is used to perform recursion? (5)
- b) Convert the expression  $((A + B) * C - (D * E) \wedge (F + G))$  to equivalent Prefix and Postfix notations? (5)
- c) What is stack? Give the codes of Push and Pop operation in stack? (5)





**Question 6:** Multi-programming (or multi-tasking) enables more than a single process to apparently execute simultaneously.

- a) How is this achieved on a uniprocessor?
- b) What is the relationship between threads and processes?
- c) What is deadlock? What is starvation? How do they differ from each other?

**Question 7:**

- a) In Java, How to make an object completely encapsulated? How is polymorphism achieved in java?
- b) What is the difference between declaration and definition?
- c) What is a Constructor? Explain it with an example.

(5)  
(5)  
(5)

**Question 8:**

- a) Consider the following output.  
1 1 1 1  
2 2 2 2  
3 3 3  
4 4  
5

Write down the C language program code for specific output?

- b) What is data dictionary?
- c) Briefly define top-down and bottom-up design model?

(5)  
(5)  
(5)





# BALUCHISTAN PUBLIC SERVICE COMMISSION

Written Evaluation Test for the Post(s) of

**Lecturer (B-17)**

In the Balochistan Agriculture College, Quetta

## COMPUTER SCIENCE

Roll Number

11

Total Marks: 100

November 7, 2017 at 09:30 AM

Time Allowed: 01 Hour

- NOTE:
- Attempt 8 in all questions, Section-I is compulsory.
  - All the parts (if any) of each Question must be attempted at one place instead of at different places.
  - Candidate must write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
  - No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
  - Extra attempt of any question or any part of the attempted question will not be considered.

### Question I:

#### A) Fill in the blank: (10)

- \_\_\_\_\_ is approximately 1,000 bytes.
- \_\_\_\_\_ is the science that attempts to create machines that will emulate the human thought process.
- You would use \_\_\_\_\_ software to create spreadsheets, type documents, and edit photos.
- When you connect to the \_\_\_\_\_, your computer is communicating with a server at your Internet service provider (ISP).
- Input devices are used to provide the steps and tasks the computer needs to process data, and these steps and tasks are called \_\_\_\_\_.
- A data structure is a logical method of representing \_\_\_\_\_.
- Process known as \_\_\_\_\_ tracks trends and allows retailers to respond to consumer buying patterns.
- The main attribute of Spiral model for software development is \_\_\_\_\_.
- The keyword used to transfer control from a function back to the calling function is \_\_\_\_\_.
- System software are categorized in \_\_\_\_\_ and \_\_\_\_\_.

#### B) For whom the following abbreviation stands for: (10)

- FORTAN
- BMP
- PHP
- SAP
- PDF
- SOAP
- VLSI
- Wi-Fi
- ASP
- UDP

#### C) Write "TRUE" or "FALSE" (10)

- When an element needs to be removed from the stack, the pop operation is performed. ....
- In a queue, elements are both inserted and deleted from one end. ....
- In a peer-to-peer network, any client computer can also be a server. ....
- A hub connects two different LANs. ....
- UTP is the cheapest cabling choice. ....
- Binary value of 303 is 0011000011. ....
- Parallel data transmission is much faster than serial transmission. ....
- Counters are digital circuit for temporary storage of data. ....
- Hexa decimal Value of 11 is C. ....
- Starting process of a computer is called cold booting. ....

**Question-3:**

- a) First convert the number into Binary then solve the following arithmetic? (5)
- $683+42$
  - $783-62$
- b) Convert the given octal number to binary? (5)
- 101010
  - 101100

**Question-4:**

Write down difference between any of two.

- Data Hiding and Encapsulation (5)
- 3G and 4G (5)
- Microcontroller and microprocessor (5)

**Question-5:**

- What do you mean by preprocessing of data in data mining? (5)
- What are the main classification of SQL statements? (5)

**Question-6:**

- Convert the expression  $((A + B) * C - (D - E) ^ (F + G))$  to equivalent Prefix and Postfix notations? (5)
- What is indirect addressing? (5)

**Question-7:**

- What is a compiler and interpreter? (5)
- What are the basic functions of assembler? (5)

**Question-8:**

- How many data formats are available in Excel? Name some of them. (5)
- What is IF function in Excel? (5)

**Question-9:**

- What is difference between C and C++ ? (5)
- What are C++ inline functions? (5)

**Question-10:**

- Name the two sub-layers of Data link layer. Specify their protocols? (5)
- What are the various types of key used in cryptography? (5)

**Question-11:**

- Describe the three levels of data abstraction? (5)
- Briefly define top-down and bottom-up design model? (5)





# PHOTO STATE

**Balochistan Public Service Commission**  
Samungli Road, Quetta Cantt.  
**WRITTEN EVALUATION TEST FOR THE POST(S) OF LECTURER (B-17)**  
**IN THE EDUCATION DEPARTMENT**

## COMPUTER I.T

Marks: 80

June 07, 2016

Time Allowed: 2 Hours 40 Minutes

**Note:** Attempt Four questions, selecting at least ONE question from each Section.  
All questions carry equal marks.

### PART-II

#### Section-A

- Q.1. (a) What is distributed system? Briefly define reasons for building such systems. (5)  
(b) Briefly describe Real Time and Time Sharing Systems. What are the constraints in designing Real Time Systems. (5)  
(c) What are the main functions of Operating Systems? (5)  
(d) What is disk scheduling? In your opinion which scheduling algorithm is the best? (5)
- Q.2. (a) Describe Instruction Execution Cycle with the help of a diagram. (5)  
(b) CPU and memory are normally connected by three groups of connections, each called a bus: Data bus, Address bus and control bus. Briefly describe the functions and communication of three buses. (5)  
(c) Differentiate between CISC and RISC Computers. (5)  
(d) What are flag registers? Briefly describe the functions of main flag registers. (5)

#### Section-B

- Q.3. (a) What are Web Applications? Briefly describe the characteristics which differentiate Web Applications from other software. (5)  
(b) What is Prototyping Model? In what circumstances this model is used to develop the software? (5)  
(c) What is Software Quality? What are the key quality attributes of computer software? (5)  
(d) Write a program in any language to generate Fibonacci Series. (5)
- Q.4. ✓(a) Define class and object how objects are inherited in OOP? (5)  
(b) Construct a Finite Automata on  $\Sigma = \{a, b\}$  which accepts only  $\epsilon$  (5)  
✓(c) What are Abstraction and Encapsulation with respect to OOP? (5)  
(d) Write an algorithm for finding an element in an array of N elements using binary search. (5)

#### Section-C

- Q.5. (a) What is data integrity? How it is maintained in DBMS? (5)  
(b) What is a sub-query? Why it is used? (5)  
(c) Briefly discuss advantages of Database Approach over File Processing System. (5)  
(d) What is difference between Homogenous and Heterogeneous database models. (5)
- Q.6. (a) What are Network Operating Systems (NOS)? Briefly explain types of Network Operating Systems. (5)  
(b) What are connectionless and connection oriented networks? (5)  
(c) What are Asynchronous and Synchronous transmission? (5)  
(d) Every device made by any company in any country has a different MAC address, Discuss briefly. (5)

*Good Luck*

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# Balochistan Public Service Commission

## WRITTEN EVALUATION TEST FOR THE POST(S) OF LECTURER (B-17) IN THE EDUCATION DEPARTMENT COMPUTER I.T.

Marks: 100

June 04, 2015 (02.30 PM)

Time Allowed: 02 Hours

Note: Attempt Five questions. Question No. 1 is Compulsory. Attempt only FOUR questions from PART-II, selecting at least ONE question from each Section. All questions carry equal marks.

### PART-I (20 Marks)

- Q-1. (A) For whom the following abbreviations stand for: (5)  
(i) ICMP, (ii) RADD, (iii) GSM, (iv) CASE, (v) SRS
- (B) Fill in the blanks with the most suitable option for the following statements: (5)  
(i) ..... is the opposite of amplification.  
(ii) ..... software helps you carry out tasks, such as typing a document or creating a spreadsheet.  
(iii) A register in the microprocessor that keeps track of the answer or results of any arithmetic or logic operation is called .....  
(iv) The decimal equivalent of hex  $(100)_{16} = (\dots)_{10}$ .  
(v) Execution of two and more programs in waiting time of each other is called .....
- (C) Write "TRUE" or "FALSE" (5)  
(i) The language understood by the microprocessor is called assembly language.  
(ii) An address bus is also called a unidirectional bus.  
(iii) A program in execution is called a process.  
(iv) An aperiodic signal consists of a continuously repeated pattern.  
(v) A domain is the set of allowable values for one or more attributes.
- (D) Define the following terms (5)  
(i) Micro blogging (ii) Selfie (iii) Mail Merge (iv) Computer Worm (v) IP version 6

### PART-II (80 Marks)

#### Section-A

- Q-2. (a) Define latency, transfer and seek time with respect to disk I/O? (5)  
(b) What is an interrupt? How are multiple interrupts dealt with? (5)  
(c) What are the steps performed by an OS to create a new process? (5)  
(d) What is the difference between a page, a frame and a segment in OS with respect to memory management? (5)
- Q-3. (a) What are the four main functions of a computer? (5)  
(b) List and briefly define the main structural components of a processor? (5)  
(c) Explain sequential and random access methods of accessing units of data in memory? (5)  
(d) What is the distinction between spatial locality and temporal locality? (5)

#### Section-B

- Q-4. (a) Briefly describe spiral model of software development with diagram? (5)  
(b) Differentiate between verification and validation with respect to software testing? (5)  
(c) Discuss LOC cost estimation technique used in software engineering? (5)  
(d) Explain two characteristics involved in software risks? (5)
- Q-5. (a) An object has state, behavior, and identity. Explain? (5)  
(b) Explain the term inheritance in object oriented programming with its types? (5)  
(c) Shortly define polymorphism in object-oriented programming with an example? (5)  
(d) Explain the following types of data structure by giving example. (i) Stack (ii) Queue (5)

#### Section-C

- Q-6. (a) A Relational Model is commonly used database model. Identify essential characteristics/properties of the Relations? (5)  
(b) What is data independence in DBMS? (5)  
(c) What is normalization? Briefly explain 3rd normal form using any database example? (5)  
(d) Define concurrency control techniques used in databases? (5)
- Q-7. (a) Explain the functions/responsibilities of the Transport Layer of OSI model? (5)  
(b) Discuss three types of errors found in data communication? (5)  
(c) Define the term Topology and explain Tree Topology with the help of a diagram? (5)  
(d) Briefly explain CSMA/CD access mechanism used in an Ethernet? (5)

Good Luck

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